4. It withstands the vibrations of the vehicle when the vehicle is going on the bumps and pot holes.

5.8 Tyre construction

It consists of the following four parts. They are,

1. The bead region
2. The side wall region
3. The shoulder region
4. The tread region

5.8.1 Bead

Bead is a part of the tyre which is used to fit the tyre on the wheel rim. Bead is made of a circular metal wire placed on both sides of the tyre. In the tyre beads one or more metal wires are covered strongly by the thick nylon or rayon threads around the metal wires. Tyre beads help to keep the tyre shape unchanged.

5.8.2 Side wall region

The side wall of the tyre is constructed to be very flexible. The vertical area between the tyre bead and tyre tread region is called as the side wall region. Wheel tube is fitted inside this region. The tyre marks are made in this side wall area only.

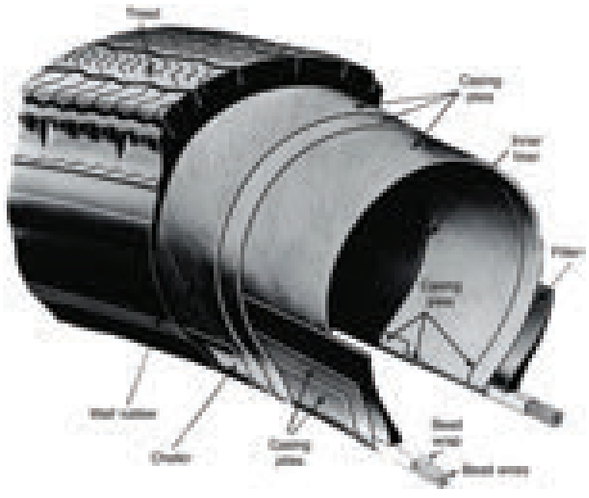


Figure 5.8 Tyre construction

5.8.3 Shoulder region

The horizontal area of region connecting the side walls and the tread is called as the shoulder region. This shoulder region creates a slightly curved look for the tyre area.

5.8.4 Tread region

The top area on the periphery of the wheel that is in contact on the road surface is called as the tread region. For giving good grip on the road, there are different shapes of cross grooves available on this region.

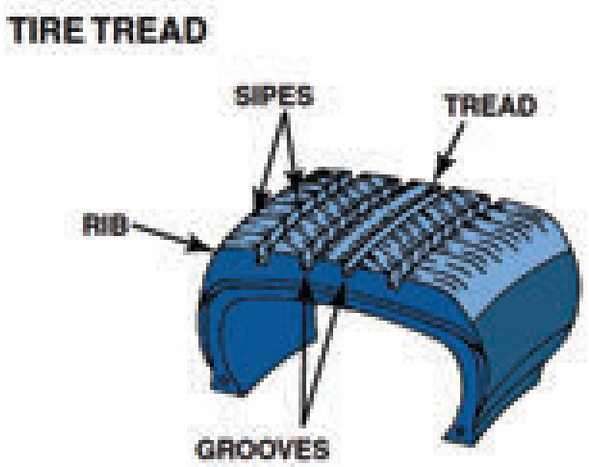


Figure 5.8.4 Tread pattern of a typical tyre

5.9 Tread types

The tread region present in the tyres is of three types. They are

1. Stud type tread
2. Symmetrical type tread
3. Winter type tread

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